Three-axis purity certificate

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2. Model and population constraints

Use output-first vectorization on $\mathcal{H} = \mathbb{C}^4 \otimes \mathbb{C}^4$, with normalized Choi vectors $|M\rangle = \mathrm{vec}(M)/2$. Let

$$p = |I_4\rangle, \quad u = |i(Z \otimes I_2)\rangle, \quad v = |i(X \otimes I_2)\rangle,$$

where X, Y, Z are the Pauli matrices. Put

$$a = sh, \quad k = \frac{(a - \sqrt{aq_0})^2}{sh(1-h)}, \quad d = s + k, \quad r = 1 - d,$$

$$\eta = \frac{\sqrt{s}\,p - \sqrt{k}\,u}{\sqrt{d}}, \qquad \theta = \frac{\sqrt{k}\,p + \sqrt{s}\,u}{\sqrt{d}}, \qquad F = |v\rangle\langle v|,$$

and $D = I - |p\rangle\langle p| - |u\rangle\langle u| - F$, of rank 13. The **exact complete lower-endpoint phase curve** with its fixed effect forces the orthogonal vector θ into the kernel of each compatible positive Choi state. For the additional-outcome probability $f = \mathrm{Tr}(F\rho)$, the fixed populations are $d, f, r - f$. The normalized CPTP condition is $\mathrm{Tr}_{\mathrm{out}}\,\rho = I_4/4$.

The minimum-purity state in this compatible class is

$$\rho_\star(f) = d|\eta\rangle\langle\eta| + fF + \frac{r-f}{13}D, \qquad J(f) = d^2 + f^2 + \frac{(r-f)^2}{13}.$$

All vectors spanning the three sectors can be chosen as maximally entangled Choi vectors; hence $\mathrm{Tr}_{\mathrm{out}}\,\rho_\star = I_4/4$. For any compatible ρ, the populations give the exact identity

$$\mathrm{Tr}\,\rho^2 = J(f) + \|\rho - \rho_\star(f)\|_2^2.$$

This setup is restricted to that exact phase curve and detector. Ordinary readout probability alone does not fix θ as a null direction.

3. Joint first-qubit X/Y/Z measurement theorem

Theorem (three-axis locally sharp purity certificate). Fix the model of section 2, with exact s, h, q_0, the full specified phase curve, and the extra-outcome population $0 \le f \le r$. Assume a normalized CPTP Choi state ρ transfers unchanged to a fresh product input $|00\rangle$. For each $Q \in \{X, Y, Z\}$, denote by p_Q the probability that a projective measurement of the **first output qubit** in the corresponding Pauli basis returns the positive eigenvalue. Set

$$p_Z^\star(f) = d + \frac{6(r-f)}{13}, \qquad p_X^\star = p_Y^\star = \tfrac{1}{2}.$$

Then

$$\boxed{\mathrm{Tr}(\rho^2) \;\ge\; J(f) + \frac{91}{1230}\big[p_Z - p_Z^\star(f)\big]^2 + \frac{7}{97}\Big[(p_X - \tfrac{1}{2})^2 + (p_Y - \tfrac{1}{2})^2\Big].}$$

The inequality holds globally over all compatible normalized CPTP Choi states. At every **interior** $0 < f < r$, equality is attainable for every sufficiently small triple of deviations from the indicated reference probabilities, using an explicitly positive normalized CPTP Choi matrix. This is *local* simultaneous sharpness; it does not assert attainability of arbitrary triples far from the reference probabilities, nor impose unital or mixed-unitary constraints.

Proof. Define the actual output-probability observables

$$M_Q = 4\,[((I_2 + Q)/2) \otimes I_2]_{\text{out}} \otimes [|00\rangle\langle 00|]_{\text{in}}.$$

Let $B = I - |\theta\rangle\langle\theta|$. For any Hermitian M, first remove the three fixed population directions:

$$M_1 = BMB - \text{Tr}(BMB|\eta\rangle\langle\eta|)|\eta\rangle\langle\eta| - \text{Tr}(BMBF)F - \frac{\text{Tr}(BMBD)}{13}D.$$

This operator is supported on B, orthogonal to the three fixed populations, and traceless. Put $L = \text{Tr}_{\text{out}} M_1$, a traceless Hermitian reference operator. Since θ is maximally entangled, for traceless Hermitian H, G,

$$\text{Tr}[B(I \otimes H)B(I \otimes G)] = \frac{7}{2}\,\text{Tr}(HG).$$

The operators $B(I \otimes H)B$ are already orthogonal to the three fixed populations. Therefore

$$\widehat{M} = M_1 - \frac{2}{7}B(I \otimes L)B$$

is the Hilbert–Schmidt projection onto the additional tangent condition $\text{Tr}_{\text{out}}\,\Delta = 0$. It is also orthogonal to θ, and to the three population constraints.

Direct Pauli-matrix multiplication in the specified 16-dimensional vectorization gives the following **exact** Gram matrices for the ordered observables (Z, X, Y):

$$G^{(1)}_{PQ} = \text{Tr}(M_{P,1}M_{Q,1}) = \begin{pmatrix} 276/13 & 9 & 9 \\ 9 & 49/2 & 21/2 \\ 9 & 21/2 & 49/2 \end{pmatrix},$$

$$G^{(L)}_{PQ} = \text{Tr}(L_P L_Q) = \begin{pmatrix} 27 & 63/2 & 63/2 \\ 63/2 & 149/4 & 147/4 \\ 63/2 & 147/4 & 149/4 \end{pmatrix}.$$

For example, the X/Y population subtraction is $M_{Q,1} = BM_QB - B/2$; $M_Q^2 = 4M_Q$ and $\text{Tr}\,M_Q = 8$ give $\text{Tr}\,M_{Q,1}^2 = 49/2$. Their partial traces have diagonal $7(|00\rangle\langle 00| - I_4/4)$ and an off-diagonal magnitude 1/2 with X and Y in orthogonal quadratures, giving $\text{Tr}\,L_Q^2 = 149/4$ and $\text{Tr}\,L_X L_Y = 147/4$. The entries involving Z follow by the same rank-one contractions, with $L_Z = 6(|00\rangle\langle 00| - I_4/4)$.

Subtracting the trace-preservation projection gives

$$G_{PQ} = \text{Tr}(\widehat{M}_P\widehat{M}_Q) = G^{(1)} - \frac{2}{7}G^{(L)} = \text{diag}\left(\frac{1230}{91}, \frac{97}{7}, \frac{97}{7}\right).$$

For $\Delta = \rho - \rho_\star(f)$, all imposed constraints are homogeneous and $\text{Tr}(M_Q\Delta) = \text{Tr}(\widehat{M}_Q\Delta)$. Orthogonal Bessel inequality in Hilbert–Schmidt space yields

$$\|\Delta\|_2^2 \geq \frac{[p_Z - p_Z^\star(f)]^2}{1230/91} + \frac{[p_X - 1/2]^2 + [p_Y - 1/2]^2}{97/7}.$$

Together with the purity-excess identity, this proves the inequality. For a sufficiently small deviation vector $(\delta_Z, \delta_X, \delta_Y)$, define

$$\rho = \rho_\star(f) + \frac{91\delta_Z}{1230}\widehat{M}_Z + \frac{7\delta_X}{97}\widehat{M}_X + \frac{7\delta_Y}{97}\widehat{M}_Y.$$

It preserves every linear phase, population and trace-preservation constraint, reproduces the desired three deviations, and achieves equality. For interior f, the base state is strictly positive on the allowed fifteen-dimensional support, so every sufficiently small Hermitian perturbation on that support remains positive. This completes the constructive local-sharpness proof. □

Restriction to the longitudinal reading. Setting $p_X = p_Y = 1/2$ reduces exactly to the coefficient $91/1230$. Measuring additional X/Y axes need not make purity identifiable: if all probabilities equal their reference values, compatible states can still have a wide range of purities.

4. Numerical witness and finite-data limits

At $f = r/2$, the verifier constructs the positive channel state with deviations $(\delta_Z, \delta_X, \delta_Y) = (0.005, 0.007, -0.004)$. It checks the full Choi marginal, positivity, phase response at six settings, exact extra-outcome probability, and equality in the three-axis bound within floating-point tolerance. It prints an attained purity of approximately 0.336263620116203.

The theorem assumes exact population parameters. For a simultaneous confidence region on (f, p_X, p_Y, p_Z), a valid conservative lower purity certificate is obtained by **minimizing the right-hand side of the theorem over that entire region**, together with any independently justified region for (s, h, q_0). For known f and marginal probability intervals, this is achieved by substituting each interval's distance from its corresponding reference probability. For uncertain f, minimize over its allowed range; the objective is convex quadratic in f when the other inputs are fixed. An empirical claim requires calibration transfer, finite-sample uncertainty allocation, and an independently supported full-phase compatibility premise. Merely inserting observed frequencies into exact equations provides no coverage guarantee.

The mutually noncommuting X, Y, Z output observables would generally be measured on different preparations. The normalized Choi marginal applies to a **trace-preserving channel**. A history-selected trace-nonincreasing instrument branch need not have that marginal, and applying the same process to product input $|00\rangle$ is an additional physical transfer claim. Nothing here verifies those premises experimentally.